

Smartphones are probably one of the first objects where us mortals got to experience augmented reality; after all they are omnipresent. However, with time, as we look around us and observe closely, we find that augmented reality is prevalent elsewhere too.

Let us have a look at the types of AR. Before we begin, understand that lines are blurred, broken and twisted. Definitions explained here are not written in stone.

Projection based AR

Just like anything else which is beyond our reach, projection based AR feels more attractive (at least as of now) compared to an AR app you can install on your phone. As is obvious by its name, projection based AR functions using projection onto objects. What makes it interesting is the wide array

of possibilities.



Projection-based AR can build you a castle in air, or a dialer on hand

One of the simplest is projection of light on a surface. Speaking of lights, surfaces and AR, did you ever think those lines on your fingers (which divide each finger into three parts) can create 12 buttons? Have a look at the image and you would quickly grasp what we're talking about. The picture depicts one of the simplest uses of projection based

AR where light is fired onto a surface and the interaction is done by touching the projected surface with hand. The detection of where the user has touched the surface is done by differentiating between an expected (or known) projection image and the projection altered by interference of user's hand.

One of the widespread uses of projection based AR techniques is non-interactive. Projection on objects can be used to create deception about the position, orientation and depth of an object. In such a case an object is taken into consideration and its structure is studied in depth. The object's distance from the projection is calculated and the projection light sequence is then designed carefully to deceive the viewer's mind.

The most exciting implementation of projection-based AR is perhaps the most exciting one of all uses. We are sure it can appeal to your inner scientist